

10-3 **D** For the circuit of Figure P10-3:

- Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.
- Select values of R and C to locate a pole at -640 rad/s. Where is the resulting zero?

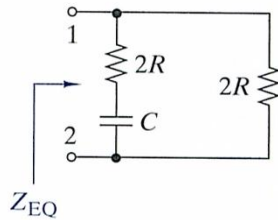


FIGURE P10-3

10-5 **D** For the circuit of Figure P10-5:

- Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.
- If $R = 2$ k Ω and $C = 0.1$ μ F, select a value of L to locate zeros at $\pm j5000$ rad/s.
- Where are the poles located once you have selected the inductor in part (b)?

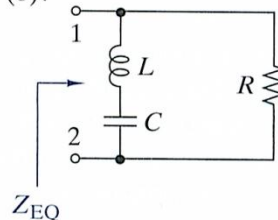
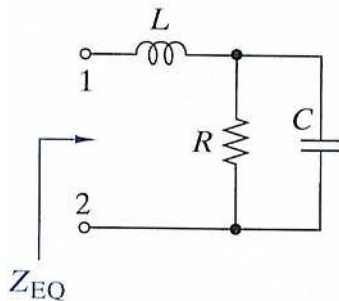


FIGURE P10-5

10-9 For the circuit of Figure P10-9:

- If $R = 1$ k Ω , $L = 2$ H, and $C = 0.5$ μ F locate the poles and zeros of $Z_{EQ}(s)$?
- If we were to increase the inductance to 5 H, how would the poles and zeros change?



10-17 **D** Find the Thévenin equivalent for the circuit in Figure P10-17. Then select values for R and L so that the Thévenin voltage has a pole at -12 krad/s.

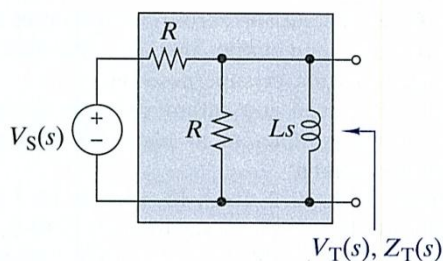


FIGURE P10-17

10-19 The switch in Figure P10-19 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$, $i_L(t)$, $V_O(s)$, and $v_O(t)$ in symbolic form.

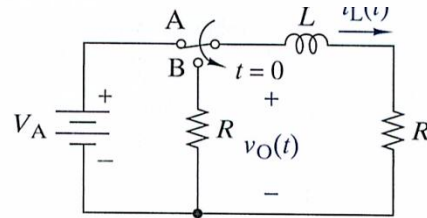


FIGURE P10-19

10-25 The switch in Figure P10-25 has been in position A for a long time and is moved to position B at $t = 0$.

- Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.
- Repeat part (a) using MATLAB.
- Find $i_L(t)$ for $R_1 = R_2 = 500$ Ω , $R_3 = 1$ k Ω , $L = 500$ mH, $C = 0.2$ μ F, and $V_A = 15$ V.

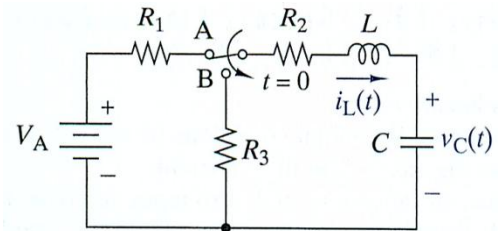


FIGURE P10-25

10-27 The circuit in Figure P10-27 is in the zero state. The s -domain relationship between the input $I_1(s)$ and the output $I_R(s)$ is usually given as a ratio called a *network function*. Find $I_R(s)/I_1(s)$. Identify the poles and the zeros.

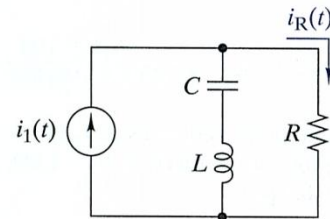


FIGURE P10-27

10-32 There is no energy stored in the capacitor in Figure P10-32 at $t = 0$. Transform the circuit into the s domain and use current division to find $v_O(t)$ when the input is $i_S(t) = 5e^{-500t}u(t)$ mA. Identify the forced and natural poles in $V_O(s)$.

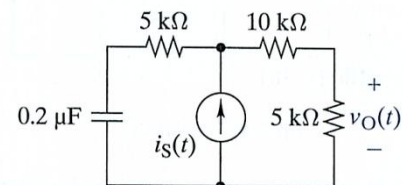


FIGURE P10-32

10-61  With the circuit in the zero state, the input to the integrator shown in Figure P10-61 is $v_1(t) = \cos 1000t$ V. The desired output is $v_2(t) = -\sin 1000t$ V. Use Laplace to select values of R and C to produce the desired output. If the capacitor had 10 V across it at $t = 0$, how would that affect the output?

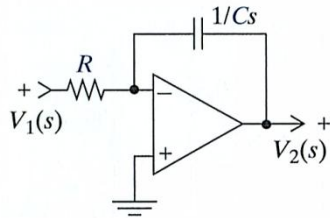


FIGURE P10-61

10-68 *s*-domain OP AMP Circuit Analysis

The OP AMP circuit in Figure P10-68 is in the zero-state. Transform the circuit into the s -domain and use the OP AMP circuit analysis techniques developed in Section 4-4 to find the relationship between the input $V_1(s)$ and the output $V_2(s)$.

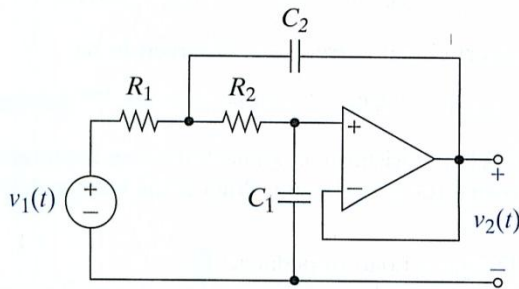


FIGURE P10-68